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HW Set 03: Numerical Integration

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ABSTRACT

The third homework: it's numerical integration time. All code can be found in the Jupyter Notebook: https://github.com/kparrine/Computational_Astro/blob/main/C3/C3.ipynb.

1. QUESTION 1

I'm making a code to solve

$$\frac{dy}{dx} = y^2 + 1 \tag{1}$$

with the Euler method and Runge-Kutta 4th order (RK4) method. I plotted the solution $y = \tan(x)$ in Figure 1 for fun and the first thing I noticed is the asymptotic behavior where around 1.5, the function goes straight up and then down and then back up again. It turns out this "bad behavior" is exactly at $\pi/2$ where the tangent is undefined because the slope is infinity.

Trying to solve this equation with Euler and RK4 results at a step size of 0.015 results in Figure 2. At first glance, it is clear RK4 outperforms Euler since I can see it approximates the solution much better than Euler. This is because the Euler method is a first order method and does not handle rapidly rising functions easily, whereas RK4 is a second order method and uses a weighted average, allowing it to better estimate the solution to the function.

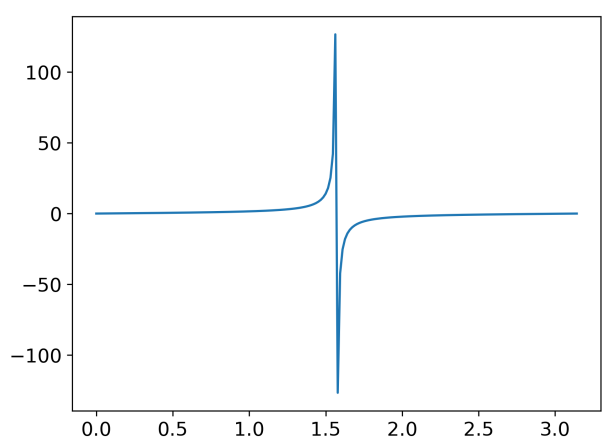
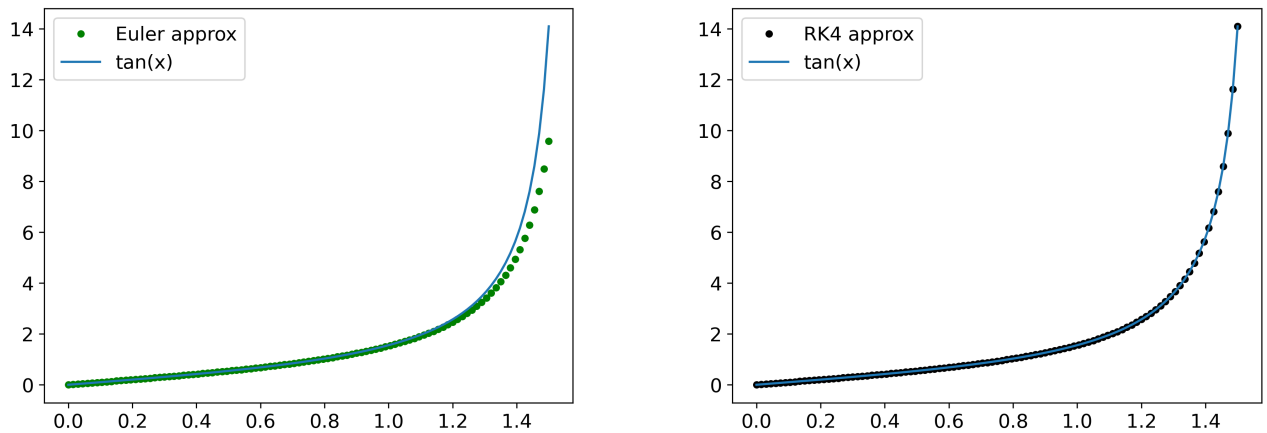


Figure 1. $\tan(x)$, the solution to Equation 1.



(a) (b)

Figure 2. (a) Euler method solution to Equation 1 and (b) RK4 solution to Equation 1.

As I said before, the slope is infinity at $\pi/2$ which causes problems for my numerical integrators such that they begin to break down. For example, if I tell my Euler function to approximate from 0 to 2, it fails because y^2 becomes too big, however, it manages to get to $x = 1.7$ as seen in Figure 3. We can see that Euler attempts to approximate the function past this "break point" and it shoots up to a y-value of 2.09×10^{164} at $x = 1.7$. This is clearly incorrect as $\tan(x = 1.7) \approx -7.697$.

The RK4 method also does not perform well at this "break point" and may exhibit a similar behavior to how the Euler method breaks. For example, if I tell RK4 to approximate from 0 to 5, it will break at $x = 1.65$ which is past the point of $\pi/2$ because the y values get too large. However, if I give it only a little nudge past this value, it can also attempt to approximate as seen in Figure 4. Like Euler, it approximates up to about that point and then jumps to a very high value of y, in this case, $y = 1.69 \times 10^3$ at $x = 1.57$. Again, this is incorrect since $\tan(x = 1.57) \approx -333.33$.

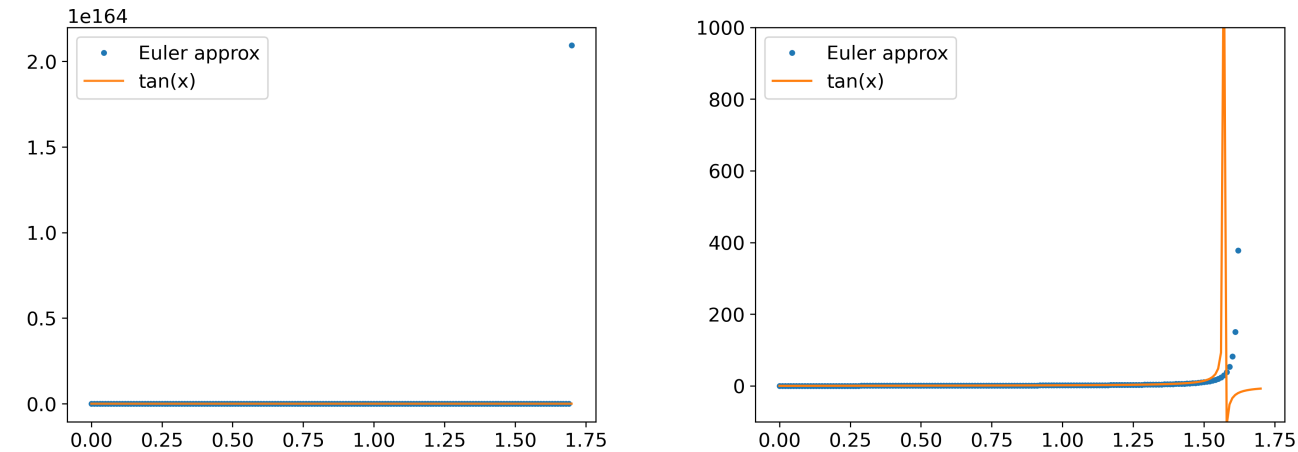
This behavior in the numerical integrators is due to the fact the $\tan(\pi/2)$ is undefined and the slope at this point is infinity.

Now, I can increase my accuracy by decreasing my step size that the function takes in each iteration. In Figure 5, I show the comparison of decreasing step sizes for both the Euler method and RK4. There is a much more noticeable difference for Euler than for RK4 with this relatively small increase in step size. The RK4 method is already quite robust and performs better than Euler, so it makes sense that Euler has a greater response to this increase in step size.

I also perform a convergence study on Euler and RK4 methods. The result is in Figure 6. The typical convergence plot is the fractional difference as a function of step size, but I have the fractional difference as a function of number of steps. My number of steps and step size are related by this equation:

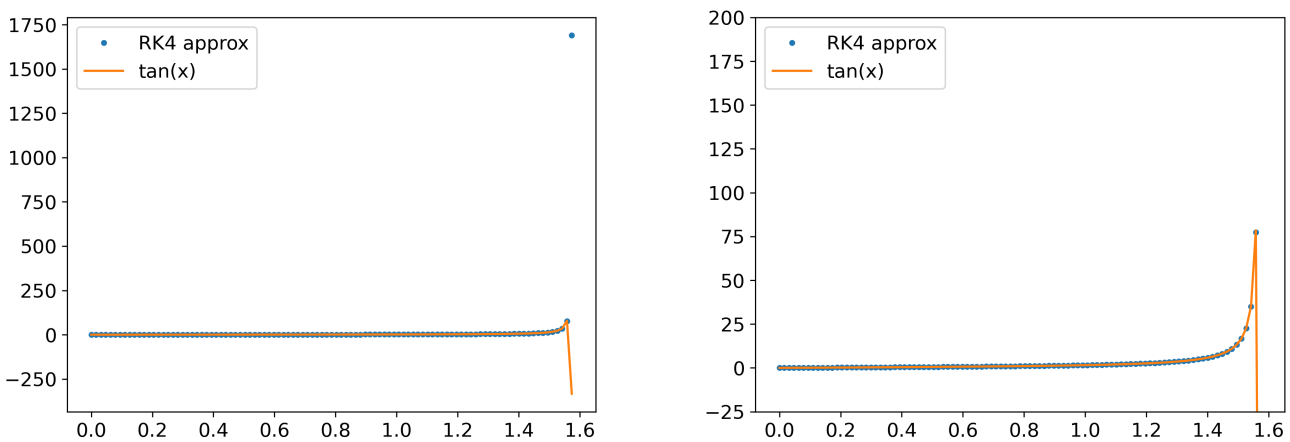
$$h = \frac{x_f - x_i}{\text{nsteps}}$$

It is clear that the fractional difference is decreasing as the number of steps increases, which agrees with the fact that as the step size decreases, the accuracy increases, therefore, this fractional difference



(a) (b)

Figure 3. Breaking the Euler method at $x = \pi/2$ (a) normal size (b) zoomed in.

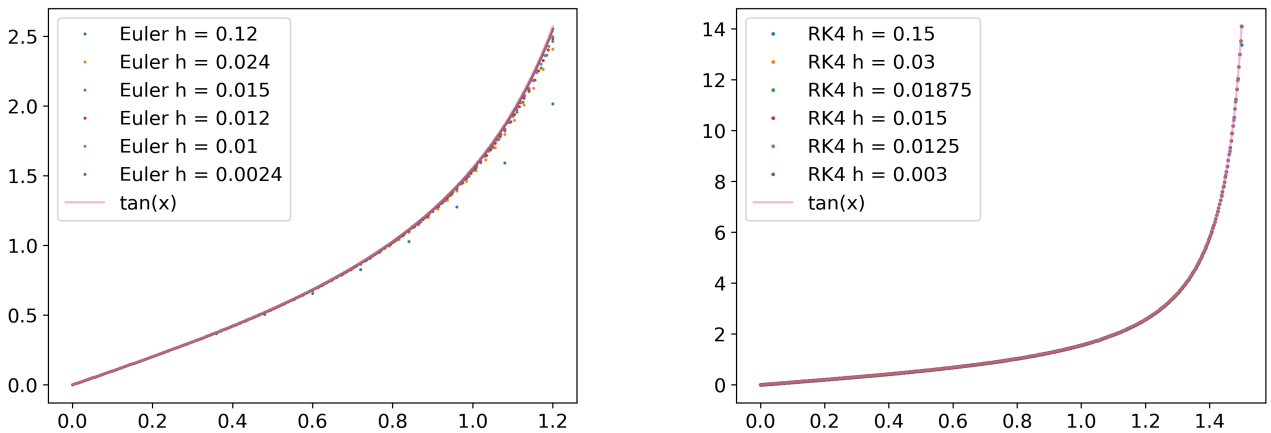


(a) (b)

Figure 4. Breaking RK4 at $x = \pi/2$ (a) normal size and (b) zoomed in.

should decrease. By eye, the slope of Euler is about 1 which agrees with the fact that it is a first order method. I do the same for RK4, and the slope is about two, again agreeing with the fact it is a second order method. However, we can see that neither of the methods have started to converge which means that I would need to increase the number of steps to achieve a true convergence. The reason why I am not showing this here is because the process is too computationally expensive on my laptop and would take many minutes to run completely.

From the convergence study, I can estimate the uncertainty of my final answer by taking the difference between the two highest-resolution cases. For Euler, my uncertainty is 5.29×10^{-3} and for RK4, my uncertainty is 3.17×10^{-11}. Even here, we see that RK4 has a much smaller uncertainty



(a) (b)

Figure 5. Increasing step size for (a) Euler and (b) RK4.

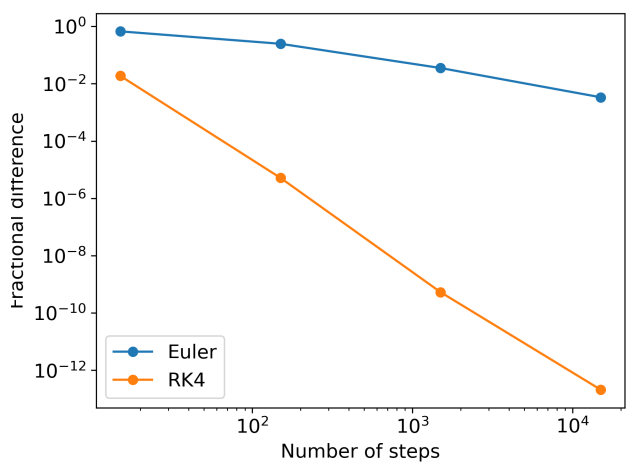


Figure 6. Convergence study for Euler (blue) and RK4 (orange).

than Euler because of its more powerful approximation. If I further increase the number of steps, these uncertainties will decrease.

2. QUESTION 2

Equation 2 is the Maxwell-Boltzmann velocity distribution.

$$f(v)dv = \left[\frac{m}{2\pi kT}\right]^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2kT}} dv \tag{2}$$

I plot this for hydrogen atoms for a 10000 K in a star for a velocity range of $[0,\ 5 \times 10^4]$ in Figure 7. I expect the most probable velocity to be where the peak of the curve is and the average velocity to be slightly towards the right of that due to the skewness of the plot. So, in Figure 7, v_p the most

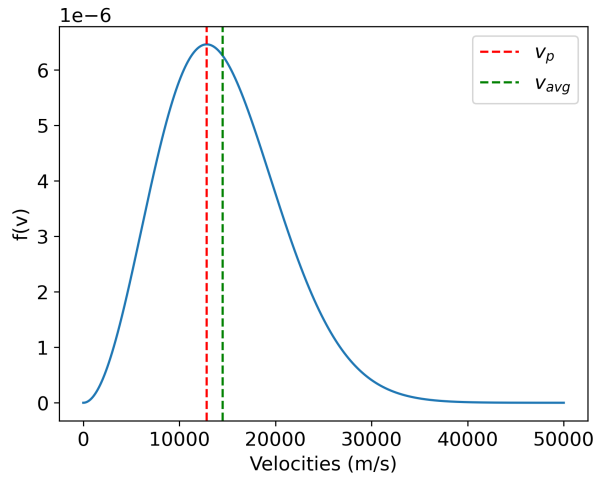


Figure 7. The Maxwell-Boltzmann velocity distribution of hydrogen atoms in a 10000 K star. The red dashed line is the most probably velocity and the green dashed line is the average velocity.

probably velocity in red falls where the peak of the curve is, and v_{avg}, the average velocity in green is towards the right, as expected. Another thing I checked for was the sum underneath the curve. Because this is a probability density, the sum under the curve should be 1. I check this and my value comes out to 0.99999880878013 which is very close. If I increase my velocity range to a higher value, this sum will get closer to one, as expected.

To calculate the fraction of Hydrogen atoms are moving fast enough for the kinetic energy to be enough to excite the atom from the ground state (n = 1) into the first excited state (n = 2), I need to find the energy, E to excite them which comes from their kinetic energy,

$$E = 1/2mv^2$$

Solving for v,

$$v_{min} = \sqrt{\frac{2E}{m}}$$

The allowed energies of the electron in the hydrogen atom are given by

$$E = \frac{-13.6 \text{ eV}}{n^2}$$

where n is the state of the electron. To find the energy to go from state 1 to state 2 we have

$$E = \left(\frac{13.6 \text{ eV}}{1^2} - \frac{13.6 \text{ eV}}{2^2}\right) = 13.6 \text{ eV} - 3.4 \text{ eV} = 10.2 \text{ eV} = 1.634 \times 10^{-18} \text{ J}$$

From this, the minimum velocity needed to excite the electrons is about 44192.64 m/s. This is the lower bound on the integral. Now I need to integrate to "infinity" with RK4. I pick my "infinity" or my upper limit to be 8×10^6 which I think is sufficiently large enough for my integrator. The fraction of hydrogen atoms that are excited from the ground state to the first excited state is 2.92×10^{-5}.

I can also look at my precision to the fraction of excited atoms with two convergence tests: 1) decreasing my step size and 2) increasing my upper limit on integration. I start with decreasing my

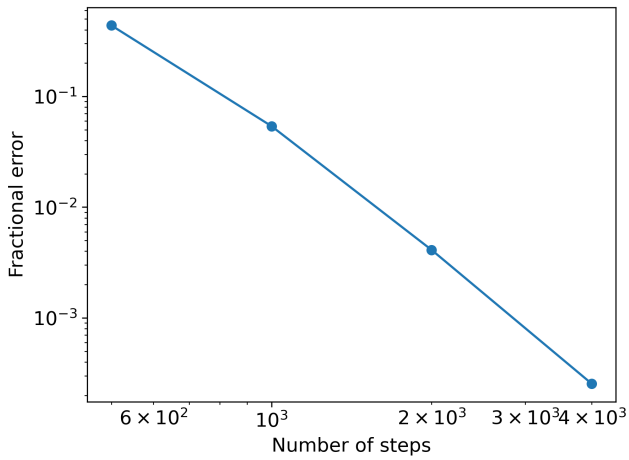


Figure 8. Convergence plot for decreasing the step size by increasing number of steps.

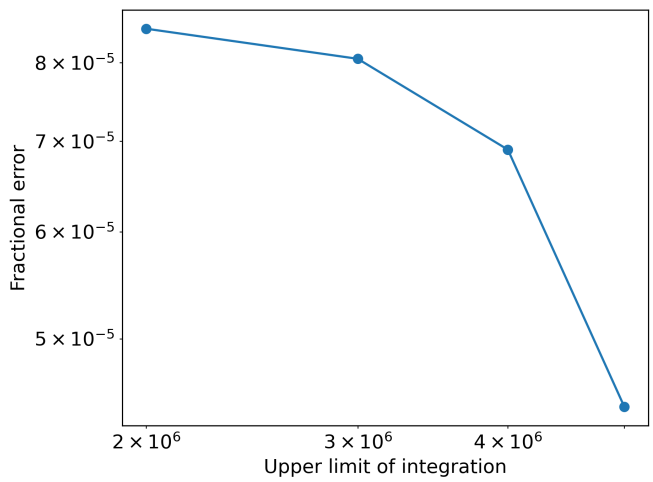


Figure 9. Convergence plot for increasing the upper limit towards infinity.

step size by increasing the number of steps. I run through a list of steps of 500, 1000, 2000, 4000, 8000, and I made a convergence plot in Figure 8. As before, I have plotted it against the number of steps and it shows the fractional difference. As expected, the fractional difference is decreasing as the number of steps increase. The uncertainty calculated is 2.54×10^{-4}. I do another convergence plot, but this time increasing the upper limit on the integral as it goes to "infinity", so I have a list of velocities that go as $2 \times 10^6, 3 \times 10^6, 4 \times 10^6, 5 \times 10^6$, and 6×10^6. In this case, I set the number of steps = 4000, to focus on how convergence reacts for just the increasing velocities. The result is Figure 9. This plot also shows a decreasing fractional difference as expected. The uncertainty from this is 4.45×10^{-5}.